Applications of linear systems

1 The total cost of 9 sharpeners and 5 sketchbooks is $\$31.55$. If the cost of a sharpener is $\$a$, and a sketchbook is $\$2.25$ more expensive. Find the value of a.

2 Maria and Buzz both walk from their houses to the bus stop every morning. Maria walks 1.5 km further than Buzz, and together they walk 3.3 km.
If Buzz walks a distance of m kilometers, find the value of m.

3 A rectangular garden bed has a perimeter of 13.2 m and the length is 3.4 m longer than the width.

a If the width of the fixture is w meters, find w to one decimal place.

b Hence, find the length of the garden bed.

4 When comparing some test results Christa noticed that the sum of her Geography test score and Science test score was 172, and that their difference was 18.
Given that her Geography score is x and her Science score is y and she scored higher for the Geography test:

a Use the sum of the test scores to form an equation.

b Use the difference of the test scores to form another equation.

c Use these two equations to find her Geography score.

d Find her Science score.

5 The number of new jobs created in Wyndburn varies greatly each year. The number of jobs created in 2012 was $260\,000$ less than triple the number of jobs created in 2007. This is equivalent to an increase of $480\,000$ jobs created from 2007 to 2012.
Let x be the number of jobs created in 2007 and let y be the number of jobs created in 2012.

a Set up two equations, in terms of. x and y, that describe the statements above.

b Use simultaneous equations to solve for x and y.

6 Luke bought some fresh produce. He picke[redacted] oranges, and 3 bananas. The cost of the Luke's shopping was $\$18.30$.
Valentina also went to the same shop and bought 5 oranges and 7 bananas. The cost of the Valentina's shopping was $\$44.03$.

- **a** If the price of oranges is represented by f and the price of bananas is represented by g, write an equation for Luke's shopping trip.
- **b** Write an equation to represent Valentina's shopping trip, then rearrange to find the value of g in terms of f.
- **c** Hence, find the price of oranges.
- **d** Find the price of bananas.

7 A mother is currently 10 times older than her son. In 3 years time, she will be 7 times older than her son.
Let x and y be the present ages of the son and mother respectively.

- **a** Use the fact that the mother is currently 10 times older than her son to set up an equation relating x and y.
- **b** Use the fact that the mother will be 7 times older than her son in 3 years time to set up another equation relating x and y.
- **c** Use simultaneous equations to solve for x and y.

8 A man is five times as old as his son. Four years ago the man was nine times as old as his son. Let x and y be the ages of the man and his son respectively.
Use simultaneous equations to solve for x and y.

9 Toby's piggy bank contains only 5 cent and 10 cent coins. If it contains 70 coins with a total value of $\$3.85$, find the number of each type of coin.
Let x and y be the number of 5 cent and 10 cent coins respectively.
Use simultaneous equations to solve for x and y.

10 Oprah invested $\$16\,000$ in total in two stocks A and B. In one year, the investment in stock A made a 14% return, while the investment in stock B fell by 6%. The total annual interest from both stocks was $\$700$.
Let x and y be the amounts, in dollars, that she invested in stocks A and B respectively.

- **a** Using the fact that she invested $\$16\,000$ in total, write an equation in terms of x and y.
- **b** Using the fact that she earned total interest of $\$700$, write an equation in terms of x and y.
- **c** Use simultaneous equations to solve for x and y.

11 Consider the diagram of the rectangle below:

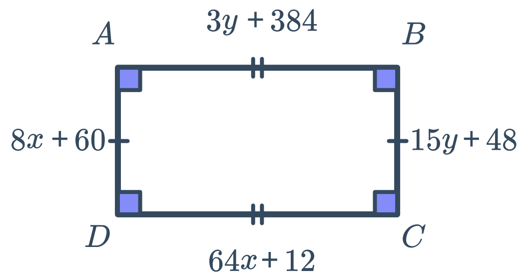
a Use the fact that $AB = CD$ to set up an equation in terms of x and y.

b Use the fact that $AD = BC$ to set up an equation in terms of x and y.

c Use simultaneous equations to solve for x and y.

d Find the length of the rectangle.

e Find the width of the rectangle.



12 Consider the following diagram of a kite:

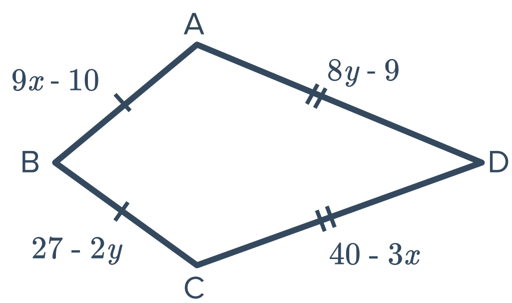
a Use the fact that $AB = BC$ to set up an equation in terms of x and y.

b Use the fact that $AD = CD$ to set up an equation in terms of x and y.

c Use simultaneous equations to solve for x and y.

d Find the length of the shorter side.

e Find the length of the longer side.



13 Consider the parallelogram below:

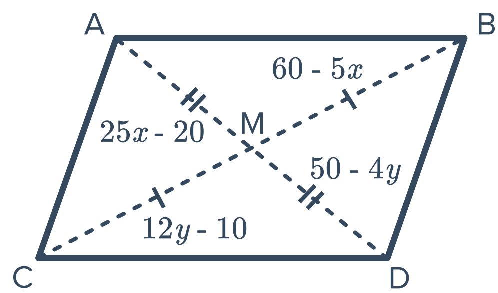
a Use the fact that $AM = DM$ to set up an equation in terms of x and y.

b Use the fact that $CM = BM$ to set up an equation in terms of x and y.

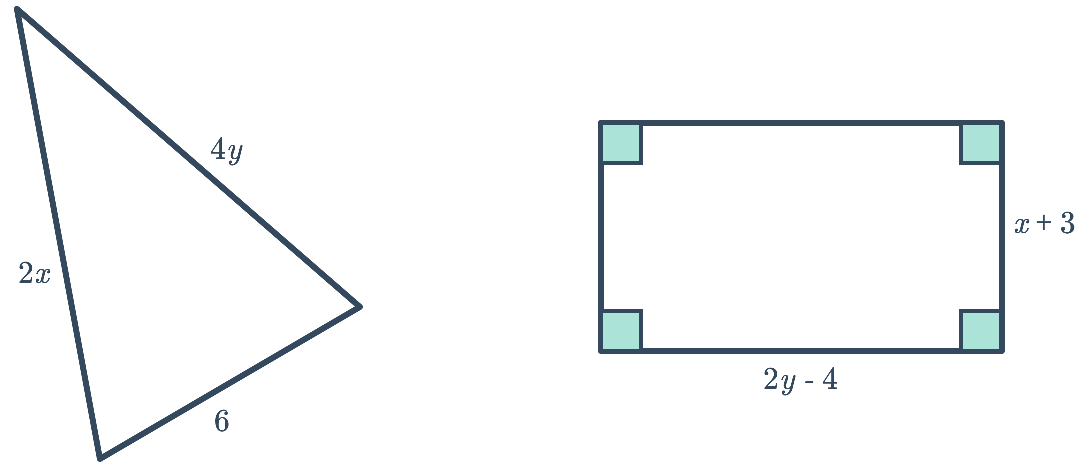
c Use simultaneous equations to solve for x and y.

d Find the length of AM.

e Find the length of side CM.



14 The perimeter of the triangle below is 56 cm[redacted] the same values for x and y are used to construct the rectangle shown. The rectangle's length is 8 cm longer than its width.



a Use the rectangle to find x in terms of y.

b Use the triangle to find another equation in terms of x and y.

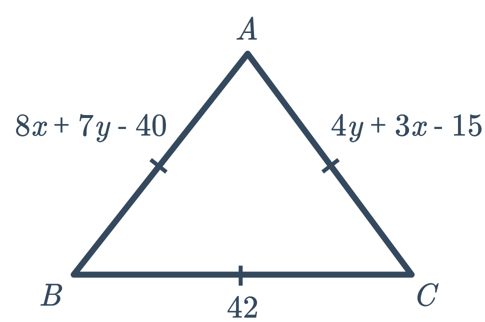
c Use simultaneous equations to solve for x and y.

15 Consider the following triangle:

a Use the fact that $AB = AC$ to set up an equation in terms of x and y.

b Use the fact that $AB = BC$ to set up an equation in terms of x and y.

c Use simultaneous equations to solve for x and y.



16 James travels in a cab on Monday and is ch[illegible] $\$5.90$. On Tuesday the return trip travelling a different route cost $\$13.90$. The fare is made up of an initial fixed fee and the kilometer meterage. The return trip was 16 km longer.

a If the initial fixed fee is $\$c$, the meterage is $\$k$ per kilometer and the trip was x kilometers on Monday, write an equation to represent Monday's trip and rearrange for c.

b Write a similar expression to represent the trip on Tuesday, then solve for k. Leave your answer to two decimal places.

c If the length of Monday's trip was 8 kilometers, solve for c. Leave your answer to two decimal places.

17 The function $f(x) = 0.47x + 8.9$ represents the U.S. annual bottled water consumption (in billions of gallons) and the function $g(x) = -0.17x + 14.2$ represents the U.S. annual soda consumption (in billions of gallons). For both functions, x is the number of years since 2009, and these functions are appropriate for the years 2009 to 2013 .

a Using the system formed by these functions, solve for x. Round your answer to the nearest whole number.

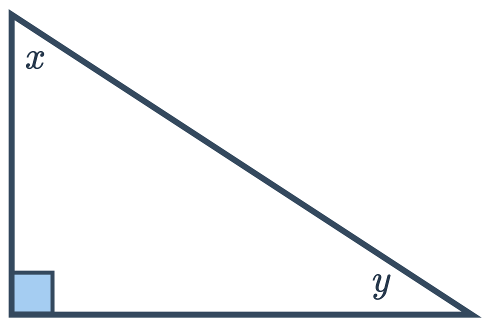
b Hence, predict the year in which the bottled water and soda consumption will be the same.

18 Consider the triangle below:

a Angle x is double the size of angle y. Write an equation to represent in terms of x and y to represent this.

b Write an equation, in terms of x and y, using the angle sum of a triangle.

c Use simultaneous equations to solve for x and y.



19 Fred spent $\$55.25$ to purchase 9 flowers. He bought two different types of flowers. Rhododendron cost $\$6.45$ each and chrysanthemums cost $\$5.75$ each.
If R is the number of rhododendrons and C is the number of chrysanthemums Fred bought:

a Write an equation in terms of R and C describing the total number of flowers bought.

b Write an equation in terms of R and C describing the total amount spent in dollars.

c Use simultaneous equations to solve for R and C.

Additional questions

20 The sum of two mystery numbers is 4. The difference of the two numbers is -2.

- **a** Write a system of equations to represent the situation.
- **b** Sketch the two lines representing these equations on the coordinate plane.
- **c** Determine the two mystery numbers.

21 Lauren and Nicole are working on an assignment together. They have broken down the work into 9 parts of the same size. Lauren works at 2 times the speed of Nicole but has 9 pieces of work to do for another subject.

- **a** Write a system of equations to represent the situation.
- **b** Sketch the two lines representing these equations on the coordinate plane.
- **c** How many parts of the assignment did each student do?

22 Katrina has to read a book for class and only has 8 days before the test. So far, Katrina has read 90 pages. She is planning on reading 15 pages each day between now and the test. The book is 215 pages long.

- **a** Write a system of equations to represent the situation.
- **b** Sketch the two lines representing these equations on the coordinate plane.
- **c** Does Katrina finish her book before the test?

23 Create a scenario for each of the following systems where the solution is viable in terms of the scenario. Explain your answer.

- **a** $y+2x=7$
$x=3+y$
- **b** $y=-4x-27$
$2y=10x+54$
- **c** $y=3x+14$
$x+y=2$

24 Ricardo solved the following system of equations used to model wildlife populations in a wildlife sanctuary, where r represents the number of rhinos and h represents the number of hippopotami.

$$\begin{cases} 6r-4h=12 \\ 3h+0.5r=15 \end{cases}$$

- **a** Explain why the solution to the system of equations is not viable.
- **b** Explain how you might interpret the solution to the system of equations to make sense in terms of the context.

25 9 chocolate croissants and 5 guava rolls cos while 3 chocolate croissants and 10 guava rolls costs $\$69$.

- **a** Write a system of equations where x represents the number of chocolate croissants and y represents the number of guava rolls.
- **b** Solve for the price of chocolate croissants and guava rolls.

26 Preeti bought some fresh seafood. She picked up 9 oysters and 7 prawns. The cost of the Preeti's shopping was $\$30.51$. Christa went to the same shop and bought 5 oysters and 8 prawns. The cost of the Christa's shopping was $\$24.72$.

- **a** Write a system of equations where f represents the price of oysters and the g represents the price of prawns.
- **b** Find the price of oysters and prawns.

27 Pensacola high school hosts an annual badminton tournament. Student tickets cost $\$3.00$ and general admission tickets cost $\$5.50$.

- **a** If 350 tickets were sold for a total of $\$1450$, how many of each type of ticket were sold?
- **b** Explain which method you used to solve and why.

28 At Soul Food Express you can buy two orders of oxtail stew and a slice of sweet potato pie for $\$38.49$, or you can get five orders of oxtail stew and three slices of sweet potato pie for $\$99.22$.

- **a** How much is the cost of an order of oxtail stew and a slice of sweet potato pie?
- **b** Explain which method you used to solve and why.

29 Each ordered pair given is the solution to a linear system of equations. For each solution, choose one pair of variables that would make the solution *nonviable*. Explain your answer.

- **a** $(0,6)$
- **b** $(2.5,8.8)$
- **c** $(100,-2)$
- **d** $(-12,-5)$

- **i** $\begin{cases} x = \text{temperature } ({}^\circ\text{F}) \\ y = \text{number of dust particles (millions)} \end{cases}$
- **ii** $\begin{cases} c = \text{number of cats} \\ m = \text{number of mice} \end{cases}$
- **iii** $\begin{cases} l = \text{length of rectangle (ft.)} \\ w = \text{width of rectangle (ft.)} \end{cases}$
- **iv** $\begin{cases} r = \text{change in sea level (cm.)} \\ w = \text{number of Warbler birds} \end{cases}$